

Practical Solutions to Real World Problems



JavaTM Servlet & JSPTM CookbookTM

O'REILLY[®]

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Embedding Multimedia in JSPs

17.0 Introduction

Most web sites include some type of multimedia and interactive programs, such as digital videos, digital audio files, Macromedia Flash movies, and Java applets. Therefore, Java web sites often integrate this type of content with servlets and JavaServer Pages (JSPs). This chapter explains the basics of embedding multimedia in Java web components. This process involves including the `object` and `embed` tags in your component's HTML output.

A JSP is the preferred choice for combining multimedia with dynamic content, because you can make the tags that you use to embed the multimedia a part of the JSP's HTML template text. However, Recipe 17.5 also shows how to include multimedia as part of a servlet's output.



If the page containing the multimedia content does not have to include any other type of dynamic output, just use a static HTML page instead of executing JSPs and servlets. A static page typically requires fewer server resources to respond to HTML page requests.

17.1 Embedding an Applet in a JSP Using `jsp:plugin`

Problem

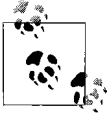
You want to use the `jsp:plugin` standard action to execute a Java applet with the Java Plug-in software.

Solution

Use the `jsp:plugin` action positioned in the area of a JSP where you want the applet to appear.

Discussion

The JSP specification provides a standard action, `jsp:plugin`, which produces the object and embed tags that are designed to allow browsers to load a Java applet. The action will run the applet using Sun Microsystems's Java Plug-in or initiate the download of the Plug-in if the user has not yet installed the Plug-in.



The Java Plug-in is designed to execute an applet using Sun Microsystems's Java 2 Runtime Environment, rather than any Java runtime provided by the browser. The installation of the Java JRE or Software Development Kit automatically installs the Java Plug-in.

Use nested `jsp:param` elements to provide the applet with any necessary parameter and value pairs. The `jsp:param` elements must be nested within a single `jsp:params` element.

Example 17-1 shows a JSP file that uses `jsp:plugin` to embed an applet named *Clock.class*. In this case, the *Clock.class* file is located in the same directory as the JSP in Example 17-1.



This applet originates from Sun Microsystems's sample applets: <http://java.sun.com/products/plugin/1.4.1/demos/plugin/applets/Clock/example1.html>

Example 17-1. Embedding a Java applet with `jsp:plugin`

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>

<jsp:useBean id="date" class="java.util.Date" />

<html>
<head><title>A Clock in a JSP</title></head>
<body>
<h2>The time...</h2>

<jsp:plugin type="applet" code="Clock.class" codebase=
    "http://localhost:8080/home/applets" jreversion="1.4.1">

<jsp:params>
    <jsp:param name="scriptable" value="false"/>
</jsp:params>

<jsp:fallback>
    Sorry, we are unable to start the Java plugin <br />
</jsp:fallback>

</jsp:plugin>

<br /><c:out value="${date}"/>
</body>
</html>
```

Users who have installed Internet Explorer for Windows depend on an HTML object tag to provide the direction for loading the applet. In browsers that support the Netscape-style plug-in, the HTML uses its `embed` tag. The `jsp:plugin` standard action generates HTML that should work with both browser types (but you still should test the resulting JSP, of course).

Example 17-2 shows the HTML tags generated by the `jsp:plugin` action when the Internet Explorer 5.5 and the Netscape browsers request the JSP in Example 17-1.

Example 17-2. HTML tags generated by the `jsp:plugin` action for loading a Java applet

```
<OBJECT classid=
  clsid:8AD9C840-044E-11D1-B3E9-00805F499D93 codebase=
  "http://java.sun.com/products/plugin/1.2.2/jinstall-1_2_2-win.cab#
  Version=1,2,2,0">

<PARAM name="java_code" value="Clock.class">

<PARAM name="java_codebase" value="http://localhost:8080/home/applets">

<PARAM name="type" value="application/x-java-applet;version=1.4.1">

<PARAM name="scriptable" value="false">

<COMMENT>

<EMBED type="application/x-java-applet;version=1.4.1" pluginspage=
  "http://java.sun.com/products/plugin/" java_code=
  "Clock.class" java_codebase=
  "http://localhost:8080/home/applets" scriptable="false"/>

<NOEMBED>
Sorry, we are unable to start the Java plugin <br />
</NOEMBED>

</COMMENT>
</OBJECT>
```

Figure 17-1 shows the JSP with the embedded applet.

See Also

The Java Plug-in technology page: <http://java.sun.com/products/plugin/>; Recipe 17.2 on embedding an applet using the Sun Microsystems HTML Converter.

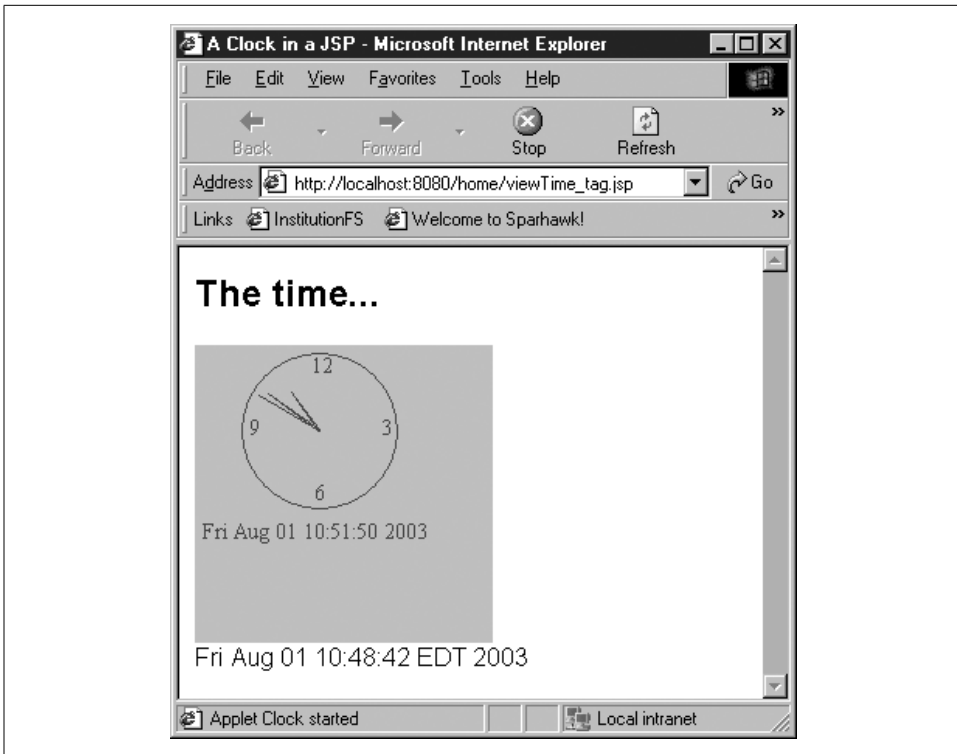


Figure 17-1. A JSP with an embedded applet

17.2 Embedding an Applet in a JSP Using the HTML Converter

Problem

You want to use the Java Plug-in HTML Converter tool to generate the tags for embedding an applet.

Solution

Use the HTML Converter tool within *htmlconverter.jar*, which is located in the *lib* directory of the directory where you have the Java SDK installed.

Discussion

A busy developer can let the Java Plug-in HTML Converter tool produce the HTML tags that are responsible for loading Java applets. The Java Plug-in is a Java-based tool that allows applets to be run in the Sun Microsystems Java 2 runtime

environment, rather than within the web browser's Java runtime environment. The Java Plug-in is installed on your machine when you install the JRE, including the installation of the SDK.

The HTML Converter tool will convert a specified JSP file that contains an applet HTML tag, replacing the applet tag with a more complex tag collection that allows most browsers to load the Java applet. The Converter leaves the rest of your JSP code untouched; it only replaces the JSP's applet tag.

Here is how to use the HTML Converter tool:

1. Write the JSP file, adding an applet tag. Example 17-3 shows a JSP that embeds a *Clock.class* applet reference. This JSP, rather redundantly, dynamically writes a time string beneath the applet. I included this code to show that the Converter does not change the JSP code; it just alters the applet tag template text included with the JSP.

Example 17-3. A JSP with an applet tag

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>

<jsp:useBean id="date" class="java.util.Date" />

<html>
<head><title>A Clock in a JSP</title></head>
<body>
<h2>The time...</h2>
<applet code="Clock.class" codebase="http://localhost:8080/home/applets">
</applet>

<br /><c:out value="${date}" />

</body>
</html>
```

2. Open a command-line window to the *lib* directory of your SDK installation, such as *H:\j2sdk1.4.1_01\lib*.
3. Type `java -jar htmlconverter.jar -gui`. This command launches the Swing version of the HTML Converter tool. Figure 17-2 shows what the tool looks like.



The HTML Converter can also be run from the command line. See the Java Plug-in Developer Guide for supported options: <http://java.sun.com/j2se/1.4.1/docs/guide/plugin/>.

4. If you want to choose a back-up folder where the tool saves the old JSP file (with the applet tag), use the HTML Converter GUI window to choose this folder.
5. Click the “Convert...” button with the JSP file specified in the top text field, and the Converter will overwrite the original file with additional object and embed tags.

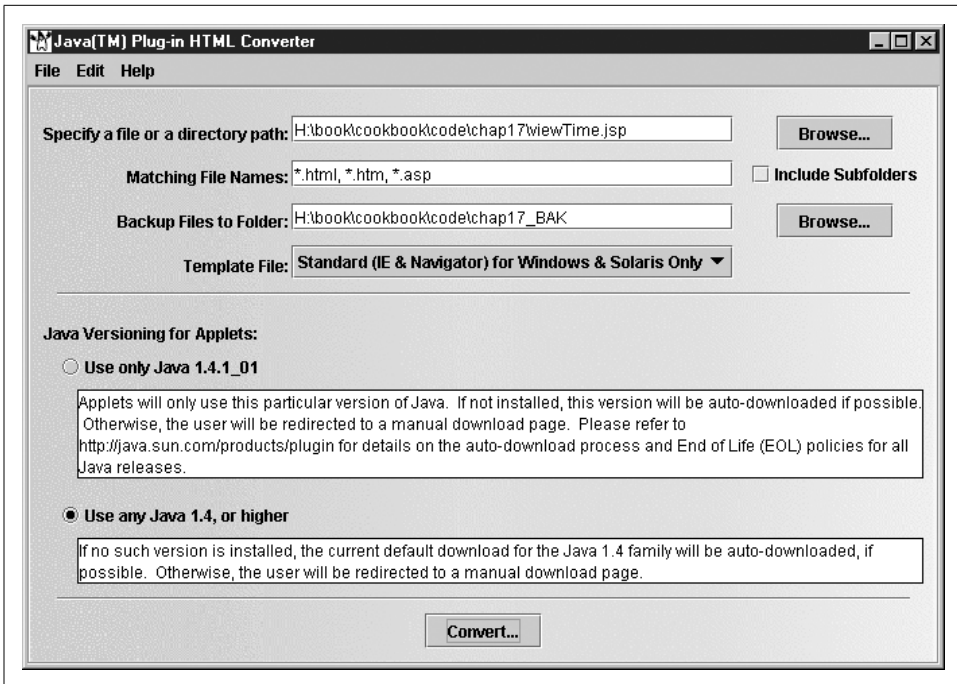


Figure 17-2. The HTML Converter (GUI version)

Example 17-4 shows the code that replaced the applet tag in Example 17-3 (in bold font), as well as the code that the converter tool did not modify.

Example 17-4. The object and embed tags produced by the HTML Converter

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>

<jsp:useBean id="date" class="java.util.Date" />

<html>
<head><title>A Clock in a JSP</title></head>
<body>
<h2>The time...</h2>

<!-- "CONVERTED_APPLET"-->
<!-- HTML CONVERTER -->
<OBJECT
  classid =
    "clsid:8AD9C840-044E-11D1-B3E9-00805F499D93"

  codebase =
    "http://java.sun.com/products/plugin/autodl/jinstall-1_4-windows-
      i586.cab#Version=1,4,0,0"
>
```

Example 17-4. The object and embed tags produced by the HTML Converter (continued)

```
<PARAM NAME = CODE VALUE = "Clock.class" >
<PARAM NAME = CODEBASE VALUE = "http://localhost:8080/home/applets" >
<PARAM NAME = "type" VALUE = "application/x-java-applet;version=1.4">
<PARAM NAME = "scriptable" VALUE = "false">

<COMMENT>
<EMBED
    type = "application/x-java-applet;version=1.4"
    CODE = "Clock.class"
    JAVA_CODEBASE = "http://localhost:8080/home/applets"
    scriptable = false
    pluginspage =
"http://java.sun.com/products/plugin/index.html#download">
    <NOEMBED>

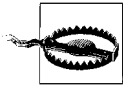
    </NOEMBED>
</EMBED>
</COMMENT>
</OBJECT>

<!--
<APPLET CODE = "Clock.class" JAVA_CODEBASE =
    "http://localhost:8080/home/applets">
</APPLET>
-->

<!-- "END_CONVERTED_APPLET" -->

<br /><c:out value="${date}"/>

</body>
</html>
```



Users may have trouble loading the applet in their browsers if they have several installed versions of the Java Plug-in. This occurs when users steadily upgrade their JRE or Java SDK versions, which install the corresponding version of the Java Plug-in. The simplest solution in these cases is to uninstall the old Java Plug-ins.

See Also

The Java Plug-in technology page: <http://java.sun.com/products/plugin/>; Recipe 17.1 on embedding a Java applet using the `jsp:plugin` standard JSP action.

17.3 Automatically Creating HTML Template for Including Flash Files

Problem

You want to automatically generate the required HTML for embedding a Flash file in a web component.

Solution

From within Macromedia Flash 6, use the “File → Publish” menu command to output an HTML file that includes the object and embed tags.

Discussion

With an *.swf* file open in Macromedia Flash 6, use the “File → Publish” menu command to create an HTML file. This file includes the necessary tags to embed the Flash movie you are working on in a web component. Then cut and paste these tags and attributes into your JSP. Example 17-5 shows the output from using this menu command with an *.swf* file named *example.swf*.

Example 17-5. Automatically generated template text from within the Flash application

```
<HTML>
<HEAD>
<meta http-equiv=Content-Type content="text/html; charset=ISO-8859-1">
<TITLE>example</TITLE>
</HEAD>
<BODY bgcolor="#FFFFFF">

<!-- URL's used in the movie-->
<!-- text used in the movie-->
<!--DeductionsPaycheckSDIAnnual SalaryMedicareSocial
securityESPP401k$#%FederalStateMarriedSingleactions-->

<OBJECT classid="clsid:D27CDB6E-AE6D-11cf-96B8-444553540000"
  codebase=
    "http://download.macromedia.com/pub/shockwave/cabs/flash/swflash.
    cab#version=6,0,0,0"WIDTH="550" HEIGHT="400" id="example" ALIGN=""
>

  <PARAM NAME=movie VALUE="example.swf">
  <PARAM NAME=quality VALUE=high> <PARAM NAME=bgcolor VALUE=#FFFFFF>

  <EMBED src="example.swf" quality=high bgcolor=#FFFFFF WIDTH="550" HEIGHT=
    "400" NAME="example" ALIGN=""
    TYPE="application/x-shockwave-flash" PLUGINSOURCE=
    "http://www.macromedia.com/go/getflashplayer">
```

Example 17-5. Automatically generated template text from within the Flash application (continued)

```
</EMBED>
</OBJECT>

</BODY>
</HTML>
```

Your JSP probably already includes the boilerplate HTML such as the body tag; therefore, you only have to cut and paste the noncommented, emphasized code in Example 17-5.



The *example.swf* file resides in the same directory as the HTML file in this example.

Example 17-6 in the next recipe shows a JSP file with the same type of Flash-related object and embed tags as those illustrated in this recipe. Figure 17-3 shows the automatically generated HTML file from Example 17-5.

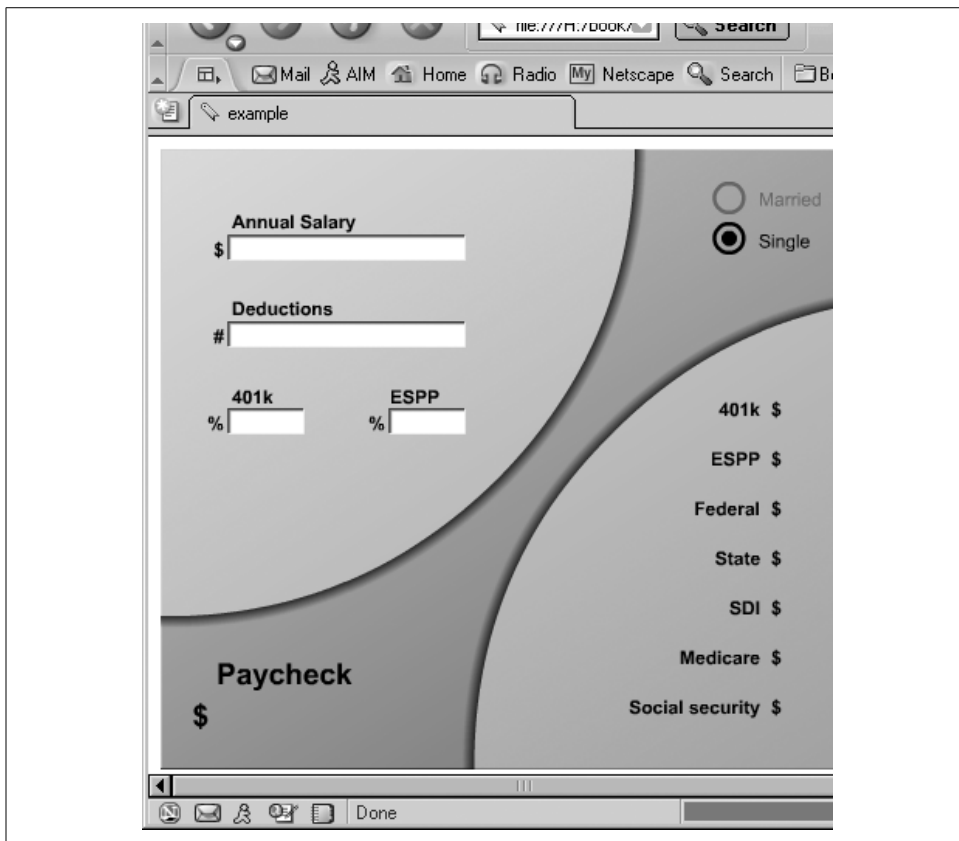


Figure 17-3. HTML template text with an embedded Flash file



The displayed Flash movie is derived from one of the Flash samples that accompanies the Flash 6 application: *Paycheck_calculator.swf*.

See Also

Macromedia technical notes page: <http://www.macromedia.com/support/flash/technotes.html>; an article about alternative techniques to using the embed tag: http://www.macromedia.com/devnet/mx/dreamweaver/articles/flash_satay.html.

17.4 Writing HTML Template to Embed a Flash File

Problem

You want to write the HTML template text to embed a Flash file in your JSP.

Solution

Use the object and embed tags so that the HTML is read correctly by the browsers that support either of these tags.

Discussion

You may not have the Macromedia Flash application that can automatically generate the HTML which is necessary to embed a Flash file (Recipe 17.4). In this case, write the required HTML template text for embedding a Flash movie inside a JSP.

Example 17-6 shows a JSP with an embedded Flash file (the embedded file has a *.swf* extension). The same concept applies to this example as to the other recipes: the object tag is for the Internet Explorer Windows browser, which embeds the media file as an ActiveX control, not a Netscape-style plug-in. The embed tag, nested inside the object tag, is designed to embed the Flash file in Netscape and other browsers that support Netscape-style plug-ins.

Example 17-6 is derived from a technical note at <http://www.macromedia.com/support/flash/technotes.html>.

Example 17-6. A JSP contains an embedded file

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>

<jsp:useBean id="date" class="java.util.Date" />

<html>
```

Example 17-6. A JSP contains an embedded file

```
<head><title>Flash in a JSP</title></head>
<body>
<h2>Enjoy the Flash Movie</h2>

<OBJECT CLASSID=
    "clsid:D27CDB6E-AE6D-11cf-96B8-444553540000" CODEBASE=
    "http://download.macromedia.com/pub/shockwave/cabs/flash/swflash.cab#
    version=6,040,0" width="293" height="423"
>

<PARAM name="movie" VALUE="coolFlashMov.swf">

<PARAM name="quality" VALUE="high">

<PARAM name="bgcolor" VALUE="#FFFFFF">

<EMBED SRC=
    "coolFlashMov.swf" quality="high" width="293" height="423"
    bgcolor="#FFFFFF" type="application/x-shockwave-flash" PLUGINSOURCE=
    "http://www.macromedia.com/go/getflashplayer"
>

</EMBED>

</OBJECT>

<br /><c:out value="${date}" />

</body>
</html>
```

Both the embed and object tags are designed to prompt the end user to download the required version of the Flash plug-in or ActiveX control if they do not already have it installed.

See Also

Macromedia technical notes page: <http://www.macromedia.com/support/flash/tech-notes.html>; an article about alternative techniques to using the embed tag: http://www.macromedia.com/devnet/mx/dreamweaver/articles/flash_satay.html.

17.5 Embedding Flash in a Servlet

Problem

You want to embed a Flash file in a servlet's output.

Solution

Use the `javax.servlet.RequestDispatcher.include(request,response)` method in the `doGet()` method of the servlet that includes the necessary HTML template text.

Discussion

The servlet can include the HTML fragment that loads the Flash movie into the page by using a `RequestDispatcher`. This process is similar to server-side includes in traditional Common Gateway Interface (CGI) programs. When the servlet receives a request, it includes the text fragment containing the Flash-related tags in its HTML output. This design separates the servlet itself from the tags and parameters that load the Flash movie, so that each of these entities evolves independently. For example, you can change the filename of the Flash movie or some of the `object` or `embed` parameters without recompiling the servlet code.

Example 17-7 is a servlet that uses a `RequestDispatcher` to include the text shown in Example 17-8. The text appears in a *flash.txt* file that is stored at the top level of the web application.



`RequestDispatchers` typically include the output of servlets and JSPs, not just text fragments. See Chapter 6 for more detailed `RequestDispatcher`-related recipes.

Example 17-7. A servlet uses a `RequestDispatcher` to include object and embed tags

```
package com.jspServletcookbook;
import javax.servlet.*;
import javax.servlet.http.*;

public class FlashServlet extends HttpServlet {

    public void doGet(HttpServletRequest request,
        HttpServletResponse response) throws ServletException,
        java.io.IOException {

        response.setContentType("text/html");
        java.io.PrintWriter out = response.getWriter();
        out.println(
            "<html><head><title>Embedded Flash content</title></head><body>");
```

Example 17-7. A servlet uses a `RequestDispatcher` to include object and embed tags (continued)

```
RequestDispatcher dispatcher = request.getRequestDispatcher(
    "/flash.txt");

dispatcher.include(request, response);

out.println("</body></html>");
} //doGet

public void doPost(HttpServletRequest request,
    HttpServletResponse response) throws ServletException,
    java.io.IOException {

    doGet(request,response);

} //doPost
}
```

Example 17-8 shows the text fragment included by the servlet in Example 17-7.

Example 17-8. An included text fragment (`flash.txt`) that a servlet uses to embed Flash

```
<OBJECT classid="clsid:D27CDB6E-AE6D-11cf-96B8-444553540000" codebase=
    "http://download.macromedia.com/pub/shockwave/cabs/flash/swflash.cab#
    version=6,0,0,0" WIDTH="550" HEIGHT="400" id="example" ALIGN=""
>

    <PARAM NAME=movie VALUE="/home/example.swf">
    <PARAM NAME=quality VALUE=high>
    <PARAM NAME=bgcolor VALUE=#FFFFFF>

    <EMBED src="/home/example.swf" quality=high bgcolor=#FFFFFF WIDTH=
        "550" HEIGHT="400" NAME="example" ALIGN="" TYPE=
        "application/x-shockwave-flash" PLUGINSOURCE=
        "http://www.macromedia.com/go/getflashplayer">
    </EMBED>

</OBJECT>
```

The result in a web browser looks exactly like Figure 17-3.

See Also

Chapter 6 on dynamically including content into servlets; Macromedia technical notes page: <http://www.macromedia.com/support/flash/technotes.html>; an article about alternative techniques to using the embed tag: http://www.macromedia.com/dev-net/mx/dreamweaver/articles/flash_satay.html.

17.6 Embedding a QuickTime Movie in a JSP

Problem

You want to embed a QuickTime movie in your JSP.

Solution

Use the `embed` tag nested inside the `object` tag. The `object` tag has to contain the `CLASSID` attribute with the proper value.

Discussion

Similar to using the Java Plug-in, a JSP uses the `embed` tag inside of an HTML `object` tag to properly load one of Apple Computer's QuickTime movies. You must include the `CLASSID` attribute value exactly as Example 17-9 specifies. You also must include the same `CODEBASE` attribute value. If the user has an Internet Explorer Windows browser, but has not yet installed the QuickTime ActiveX control, the `CODEBASE` attribute value specifies where the user can download it.

Example 17-9. Embedding a QuickTime movie in a JSP

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>

<jsp:useBean id="date" class="java.util.Date" />

<html>
<head><title>QuickTime in a JSP</title></head>
<body>
<h2>Ladies and Gentlemen, The Who</h2>

<OBJECT CLASSID=
    "clsid:02BF25D5-8C17-4B23-BC80-D3488ABDDC6B" WIDTH="320"
    HEIGHT="256" CODEBASE="http://www.apple.com/qtactivex/qtplugin.cab">

<PARAM name="SRC" VALUE="http://www.parkerriver.com/films/who_bene2.mov">

<PARAM name="AUTOPLAY" VALUE="true">

<PARAM name="CONTROLLER" VALUE="true">

<EMBED SRC=
    "http://www.parkerriver.com/films/who_bene2.mov"
    WIDTH="240" HEIGHT="196"
    AUTOPLAY="true" CONTROLLER=
    "true" PLUGINSOURCE="http://www.apple.com/quicktime/download/">

</EMBED>

</OBJECT>
```

Example 17-9. Embedding a QuickTime movie in a JSP (continued)

```
<br /><c:out value="{date}"/>

</body>
</html>
```

If the browser uses Netscape-style plug-ins, then the `embed` tag will initiate the loading of the QuickTime movie. The JSP in Example 17-6 properly loaded the movie into the Safari web browser on my Macintosh laptop, for instance. One of the advantages of the `embed` tag is that you can use a number of proprietary attributes that the embedded object, such as QuickTime, understands. Example 17-9 specifies that the movie should start playing as soon as the browser has loaded enough data (`AUTOPLAY="true"`) as well as that the browser should show the movie controls, which lets the user stop or start the movie (`CONTROLLER="true"`).

Figure 17-4 shows the QuickTime movie embedded in the JSP of Example 17-9.

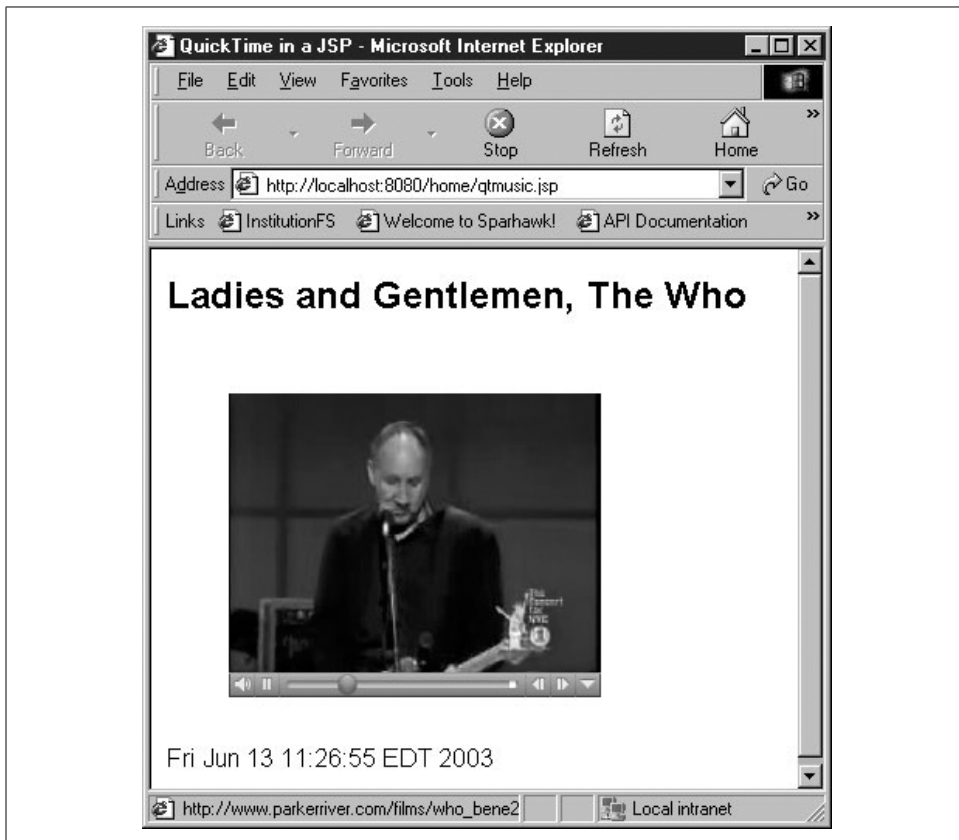


Figure 17-4. A QuickTime movie embedded in a JSP

See Also

Apple Computer's guide on embedding QuickTime in Web pages: <http://www.apple.com/quicktime/authoring/embed.html>; Recipes 17.3–17.5 on embedding a Flash file; Recipe 17.7 on embedding an SVG file in a JSP; Recipe 17.8 on embedding a background soundtrack.

17.7 Embedding an SVG File in a JSP

Problem

You want to display a Scalable Vector Graphics (SVG) image inside a JSP.

Solution

Use the `embed` HTML element to position the SVG in the JSP.

Discussion

Developers typically use the `embed` tag to place an SVG file in an HTML file or JSP. SVG is an XML-based graphics technology that provides developers and designers leverage in producing and displaying interactive graphics.



Browsers use special SVG viewer applications to handle the embedded SVG files. Adobe System's SVG Viewer application can be downloaded from <http://www.adobe.com/svg/viewer/install/>. Corel's SVG Viewer can be downloaded from <http://www.corel.com/svgviewer/>.

Example 17-10 embeds an SVG file named *testLogo.svg* and points the user to the Adobe SVG Viewer download site if they have not installed an SVG Viewer application.



SVG files have extensions of either `.svg` or (in compressed form) `.svgz`, even though they are XML files.

Example 17-10. An SVG graphics file embedded in a JSP

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>
<jsp:useBean id="date" class="java.util.Date" />

<html>
<head><title>SVG in a JSP</title></head>
<body>
<h2>A Scalable Vector Graphics example</h2>
```

Example 17-10. An SVG graphics file embedded in a JSP

```
<embed src=
  '<c:out value="${param.svg_source}"/>.svg' width=
    "200" height="200" type="image/svg+xml" pluginspage=
      "http://www.adobe.com/svg/viewer/install/"
>

<br /><c:out value="${date}"/>

</body>
</html>
```

Example 17-10 shows how to place an SVG within other JSP code elements, such as the `taglib` directive and the `jsp:useBean` standard action. Example 17-10 also dynamically loads an SVG based on the request parameter named `svg_source`. The code uses the JSTL `c:out` tag and the EL's `param` implicit object to output the parameter value (see Chapter 23 on the JSTL).

Figure 17-5 shows the result of requesting the JSP in Example 17-10, including the name of the SVG file as a request parameter. The request URL looks like:

`http://localhost:8080/home/svg.jsp?svg_source=testLogo`

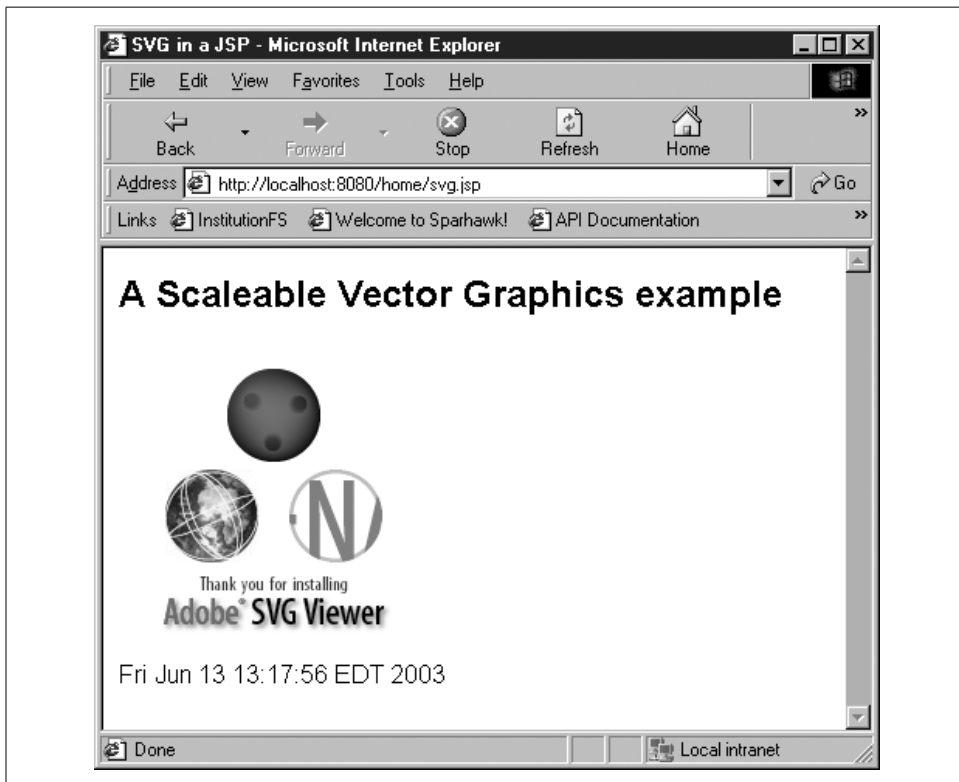


Figure 17-5. A JSP page shows an SVG graphics file

The SVG shown in Figure 17-5 is derived from Adobe Systems Inc., which creates the Adobe SVG Viewer and an SVG-enabled graphics application, Adobe Illustrator.

See Also

SVG specifications at the W3 Consortium: <http://www.w3.org/Graphics/SVG/Overview.htm>; Adobe's SVG Viewer install page: <http://www.adobe.com/svg/viewer/install/>; Recipes 17-3-5 on embedding a Flash file in servlets and JSPs; Recipe 17.6 on embedding a QuickTime movie in a JSP.

17.8 Embedding a Background Soundtrack in a JSP

Problem

You want to embed an audio file in your JSP.

Solution

Use the `embed` tag in the JSP. Use the `hidden` attribute if you want to hide the audio controls; otherwise, specify a `width` and `height` attribute for showing the audio controls.

Discussion

The `embed` tag is used to include an audio file with a JSP, so that when a user requests the JSP, the browser plays music. Specifically, the browser is designed to detect the MIME type of the embedded file, then activate a helper application such as QuickTime or RealAudio to handle the embedded file and play the music.

Example 17-11 shows a JSP that embeds an MPEG, audio layer 3 (MP3) file. The JSP displays some information about the artist based on a request parameter; this random information is included to show how to combine JSP code with the `embed` tag. The `embed` tag includes `width` and `height` attributes to show the audio controls in the web page. The controls allow the user to turn the volume off or down if they do not want to be serenaded while surfing.

Example 17-11. A JSP with an embedded audio file

```
<%@ taglib uri="http://java.sun.com/jstl/core" prefix="c" %>

<c:set var="artist" value="${param.artist}" />

<html>
<head><title>Choose Your Tunes</title></head>
<body>
```

Example 17-11. A JSP with an embedded audio file (continued)

```
<h2>You chose music from the artist <c:out value="${artist}" /></h2>

<embed src="ConstantCraving.mp3" width="240" height="160">
</embed>

</body>
</html>
```

Figure 17-6 shows the output from the JSP in Example 17-11.



Figure 17-6. Embedded song file controls in a JSP

See Also

Recipes 17.1 and 17.2 on embedding a Java applet in a JSP; Recipes 17.3–17.5 on embedding a Flash file in a JSP; Recipe 17.6 on embedding a QuickTime movie; Recipe 17.7 on embedding an SVG file.